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Markov chain Monte Carlo Significance tests

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Abstract

Markov chain Monte Carlo significance tests were first introduced by Besag and Clifford in [4]. These methods produce statistical valid p-values in problems where sampling from the null hypotheses is intractable. We give an overview of the methods of Besag and Clifford and some recent developments. A range of examples and applications are discussed.

1 Introduction

Suppose we have observed a data point X_0 belonging to some space $\mathcal{X}$. A simple statistical hypothesis asks whether X_0 was drawn from a fixed distribution π on $\mathcal{X}$. One way to test this hypothesis is by a Monte Carlo simulation. We generate a "sea" of hypothetical data points $\widetilde{X}_1, \ldots, \widetilde{X}_M$ each sampled independently according to the distribution π. The observed data X_0 is compared to the samples $\widetilde{X}_1, \ldots, \widetilde{X}_M$ which together approximate π. In the applications considered here, the space $\mathcal{X}$ will be high dimensional. Thus, comparing X_0 to $\widetilde{X}_1, \ldots, \widetilde{X}_M$ requires choosing a test statistic T that maps X_0 to some real-valued quantity of interest $T(X_0)$. If $T(X_0)$ is larger than most of $T(\widetilde{X}_1), \ldots, T(\widetilde{X}_M)$, then we have evidence against the null hypothesis $X_0 \sim \pi$.

For certain distributions π, it may be difficult or impossible to create independent and identically distributed samples $\widetilde{X}_1, \ldots, \widetilde{X}_M \overset{\text{iid}}{\sim} \pi$. For example, π may be the uniform distribution on a large combinatorial set whose elements cannot be enumerated [4, 9, 8, 12]. Or, the null distribution π may only be known up to a normalizing constant [2, 3, 16]. In [4], Besag and Clifford demonstrated how to use Markov chain Monte Carlo (MCMC) techniques to sample $\widetilde{X}_1, \ldots, \widetilde{X}_M$ and keep the statistical validity of the Monte Carlo test. Their methods place minimal assumptions on the Markov chain and only require that π is a stationary distribution of the Markov chain. The chain need not be rapidly mixing or even connected.

Here we review the theory of MCMC significance tests while emphasizing the role of exchangeability. Some interesting properties of these methods are studied through examples, and we end with a survey of applications.

1.1 Overview

The paper is structured as follows. In Section 2 we review Monte Carlo tests and discuss the validity and consistency of Monte Carlo tests. In Section 3 we show how to use MCMC to conduct valid significance tests. We describe the well-known MCMC tests introduced in [4] and a more recent generalization from [8]. We compare these three methods and discuss their relative advantages. Two examples in Section 4 give insight into how MCMC tests work, and how they differ from standard Monte Carlo tests. Finally, in Section 5 we present applications of MCMC tests to goodness of fit testing, the detection of gerrymandering and conditional independence testing.

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1.2 Notation and background

Throughout this paper $\mathcal{X}$ will be a set and $X_0 \in \mathcal{X}$ will represent observed data. The focus is on testing a hypothesis of the form $X_0 \sim \pi$ where π is a fixed distribution on $\mathcal{X}$.

A Markov chain on $\mathcal{X}$ will be described by a *transition kernel* K. That is, a sequence $X_0, X_1, \ldots, X_M$ will be a Markov chain with transition kernel K if for all $i = 1, \ldots, M$

$$X_i \mid X_0, X_1, \ldots, X_{i-1} \sim K(X_{i-1}, \cdot),$$

where $K(x, \cdot)$ is a distribution on $\mathcal{X}$ that depends on x. We will call the sequence $X_0, X_1, \ldots, X_M$ a Markov chain generated by K or a Markov chain that transitions according to K.

We will make the assumption that the null distribution π is a stationary distribution of K. That is, the distribution π is preserved by the transition kernel K. Formally, if $X_0 \sim \pi$ and $X_1 \mid X_0 \sim K(X_0, \cdot)$, then marginally $X_1 \sim \pi$ also. As discussed above, we do not assume that π is the *only* stationary distribution of K.

For simplicity, we will assume that π has a density $f : \mathcal{X} \to [0, \infty)$ with respect to some reference measure λ. We will also assume that K has a transition kernel density $k : \mathcal{X} \times \mathcal{X} \to [0, \infty)$ with respect to λ. These assumptions mean that

$$\pi(S) = \int_S f(x)\lambda(dx) \text{ and } K(x_0, S) = \int_S k(x_0, x)\lambda(dx),$$

for all $x_0 \in \mathcal{X}$ and (measurable) subsets $S \subseteq \mathcal{X}$. Under these assumptions, the condition that π is a stationary distribution for K becomes

$$f(y) = \int_{\mathcal{X}} f(x)K(x, y)\lambda(dx) \text{ for all } y \in \mathcal{X}.$$

The L-th power of the transition kernel K will be written as K^L. If $X_0, X_1, \ldots, X_{ML}$ is a Markov chain generated by K, then $X_0, X_L, X_{2L} \ldots, X_{ML}$ is a Markov chain generated by K^L. In words, L steps of a Markov chain with transition kernel K corresponds to one step of a Markov chain with transition kernel K^L. Since π is a stationary distribution of K, π is also a stationary distribution for K^L.

2 Monte Carlo tests

Monte Carlo methods are used in statistical testing when the distribution of a test statistic under the null cannot be computed analytically. A Monte Carlo significance test of the hypothesis $H : X_0 \sim \pi$ proceeds as follows:

1. Generate samples $\widetilde{X}_1, \ldots, \widetilde{X}_M \overset{\text{iid}}{\sim} \pi$ independently of X_0.
2. Calculate the test statistic $T_0 = T(X_0) \in \mathbb{R}$ and statistics $\widetilde{T}_1 = T(\widetilde{X}_1), \ldots, \widetilde{T}_M = T(\widetilde{X}_M)$ for comparison.
3. Return

$$\begin{aligned} p_{MC} &= \frac{|\{1 \le i \le M : \widetilde{T}_i \ge T_0\}| + 1}{M + 1} \\ &= \frac{\sum_{i=1}^{M} I[\widetilde{T}_i \ge T_0] + 1}{M + 1}, \end{aligned} \tag{1}$$

where $|S|$ denotes the size of a set S, and $I[S]$ is the indicator random variable for the event S.

The quantity p_{MC} will be called a *Monte Carlo p-value*. If we wish to emphasize the fact that $\widetilde{X}_1, \ldots, \widetilde{X}_M$ are i.i.d., then we will call p_{MC} the *standard* Monte Carlo p-value. Later we will discuss using MCMC to sample $\widetilde{X}_1, \ldots, \widetilde{X}_M$. More generally,

Definition 2.1. A *sampler* is a randomized algorithm that takes as input an initial point $X_0 \in \mathcal{X}$ and a required number of samples M. The output is a random sequence of samples $\widetilde{X}_1, \ldots, \widetilde{X}_M \sim P_{X_0}$.

Given a sampler, we can sample $\widetilde{X}_1, \ldots, \widetilde{X}_M \sim P_{X_0}$ and then calculate p_{MC} as above. The properties of p_{MC} will depend on the sampler. Two properties, validity and consistency, are of particular importance.

2.1 Validity

Small values of p_{MC} give statistical evidence against the null hypothesis $H : X_0 \sim \pi$. If we observe $p_{MC} \leq \alpha$, then we can reject H at significance level α. For this to be a valid test, we require a bound on the probability of falsely rejecting H. That is, when the null hypothesis holds, the probability of observing $p_{MC} \leq \alpha$ should be at most α. If this holds for all possible values of $\alpha \in [0, 1]$, then we will say that the sampler used to produce $\widetilde{X}_1, \ldots, \widetilde{X}_M$ is *valid*.

The validity of a sampler is closely related to the exchangeability of $X_0, \widetilde{X}_1, \ldots, \widetilde{X}_M$. In particular, we are interested in samplers that make $X_0, \widetilde{X}_1, \ldots, \widetilde{X}_M$ exchangeable under the null $X_0 \sim \pi$.

Definition 2.2. A sampler P_{X_0} is an *exchangeable sampler* if $X_0 \sim \pi$ and $\widetilde{X}_1, \ldots, \widetilde{X}_M \mid X_0 \sim P_{X_0}$ implies that $X_0, \widetilde{X}_1, \ldots, \widetilde{X}_M$ are exchangeable. That is, if $X_0 \sim \pi$ and $\widetilde{X}_1, \ldots, \widetilde{X}_M$ are sampled according to P_{X_0}, then the joint distribution $X_0, \widetilde{X}_1, \ldots, \widetilde{X}_M$ is invariant under all permutations.

Note that sampling $\widetilde{X}_1, \ldots, \widetilde{X}_M$ i.i.d. from π is an exchangeable sampler. The next proposition states that all exchangeable samplers are valid.

Proposition 2.3. *If $X_0 \sim \pi$ and $\widetilde{X}_1, \ldots, \widetilde{X}_M \sim P_{X_0}$ is the output of an exchangeable sampler, then*

$$\mathbb{P}(p_{MC} \leq \alpha) \leq \alpha \text{ for all } \alpha \in [0, 1]. \tag{2}$$

That is, all exchangeable samplers are valid.

Proof. Fix $\alpha \in [0, 1]$ and assume that $X_0 \sim \pi$. By definition, p_{MC} is less than α if and only if T_0 is in top α proportion of $T_0, \widetilde{T}_1, \ldots, \widetilde{T}_M$. Since $\widetilde{X}_1, \ldots, \widetilde{X}_M$ is the output of an exchangeable sampler, $X_0, \widetilde{X}_1, \ldots, \widetilde{X}_M$ are exchangeable. This implies that $T_0, \widetilde{T}_1, \ldots, \widetilde{T}_M$ are also exchangeable. By symmetry, we can conclude that the probability that T_0 is in the top α proportion of $T_0, \widetilde{T}_1, \ldots, \widetilde{T}_M$ is at most α. □

We will now study the consistency of p_{MC}.

2.2 Consistency

The standard Monte Carlo p-value is an estimate of what we will call the *analytic p-value*. The analytic p-value is the probability of observing a value of $T(\widetilde{X})$ "as extreme" as $T(X_0)$ when $\widetilde{X} \sim \pi$. The analytic p-value is thus the tail probability

$$p_A = \pi\left(\{x \in \mathcal{X} : T(x) \geq T(X_0)\}\right).$$

Suppose we treat X_0 as fixed and sample $\widetilde{X}_1, \ldots, \widetilde{X}_M$ i.i.d. from π. The indicator variables $I[\widetilde{T}_i \geq T_0]$ are then i.i.d. Bernoulli random variables with expectation p_A. Thus, p_{MC} converges in probability to p_A by the weak law of large numbers. If this occurs when we use a different sampler to produce $\widetilde{X}_1, \ldots, \widetilde{X}_M$, then we will say that the sampler is consistent.

Definition 2.4. A sampler P_{X_0} is *consistent* if for all $X_0 \in \mathcal{X}$, $\widetilde{X}_1, \ldots, \widetilde{X}_M \sim P_{X_0}$ implies

$$p_{MC} \xrightarrow{\mathbb{P}} p_A \text{ as } M \to \infty.$$

The definition of consistency is conditional on X_0. A sampler is consistent if and only if p_{MC} converges to p_A for every possible value of X_0. On the other hand, the definition of validity is marginal over X_0. The probability in (2) is defined with respect to both the randomness in $X_0 \sim \pi$ and the randomness in $\widetilde{X}_1, \ldots, \widetilde{X}_M \sim P_{X_0}$.

Moving forward, we will study samplers that use Markov chains to produce $\widetilde{X}_1, \ldots, \widetilde{X}_M$. All of these samplers will be valid but, without further assumptions, they may be inconsistent.

3 Exchangeable MCMC samplers

Suppose now that we are unable to sample directly from π and instead use MCMC to construct a sampler. The simplest approach would be *sequential MCMC*. In sequential MCMC, we sample $\widetilde{X}_1, \ldots, \widetilde{X}_M$ with a Markov chain that starts at X_0 and transitions according to K^L for some fixed value of L. If we assume the null hypothesis $X_0 \sim \pi$, then we will have $\widetilde{X}_i \sim \pi$ for every i since π is a stationary distribution of K. This implies that $\widetilde{X}_1, \ldots, \widetilde{X}_M$ each have the same marginal distribution as X_0, but it does not imply that $X_0, \widetilde{X}_1, \ldots, \widetilde{X}_M$ are exchangeable. Indeed, the joint distribution of $X_0, \widetilde{X}_1, \widetilde{X}_2$ will typically be different to the joint distribution of $X_0, \widetilde{X}_2, \widetilde{X}_1$.

The lack of exchangeability means that we can not apply Proposition 2.3. Thus, sequential MCMC may be an invalid sampler. There are natural examples where $\mathbb{P}(p_{MC} \leq \alpha)$ is of the order $\sqrt{\alpha}$ when $X_0 \sim \pi$ and sequential MCMC is used. Section 6 of the supplementary material of [9] gives details about one such example.

The following MCMC methods are exchangeable samplers. They achieve exchangeability by using both the transition kernel K and a second transition kernel $\widehat{K}$ called the reversal of K.

Definition 3.1. Suppose that π is a stationary distribution of K. A *reversal of* K is any transition kernel $\widehat{K}$ with transition kernel density $\widehat{k}$ such that for all $x, y \in \mathcal{X}$

$$f(x)k(x,y) = f(y)\widehat{k}(y,x),$$

where f is the density of π and k is the kernel density of K. If $\widehat{K} = K$, then K is said to be *reversible*.

Remark. If $f(x) > 0$ for all $x \in \mathcal{X}$, then K has a unique reversal with respect to π with kernel density given by

$$\widehat{k}(y,x) = \frac{f(x)}{f(y)}k(x,y).$$

We will thus often refer to $\widehat{K}$ as *the* reversal of K.

The name reversal is justified by the following proposition (see, for example, Chapter 2.4.2 in [5]).

Proposition 3.2. *Suppose that $X_0, X_1, \ldots, X_M$ is a Markov chain with transition kernel $\widehat{K}$ and initial distribution $X_0 \sim \pi$. The time-reversed sequence $X_M, X_{M-1}, \ldots, X_0$ has the same distribution as a Markov chain with transition kernel K and initial distribution $X_M \sim \pi$.*

Equipped with Definition 3.1, we are now ready to describe and study MCMC exchangeable samplers.

3.1 The parallel method

Our first MCMC exchangeable sampler is called the *parallel method* [4]. It proceeds as follows:

1. Starting from X_0, run a Markov chain according to $\widehat{K}$ for L steps to sample X^*.
2. Independently, for $i = 1, \ldots, M$, starting from X^*, run a Markov chain according to K for L steps to sample $\widetilde{X}_i$.

The parallel method is also called the *hub-and-spoke sampler* [2] due to the graphical representation of the method shown in Figure 1.

Proposition 3.3. *The parallel method is an exchangeable sampler.*

Proof. First note that $X_0, \widetilde{X}_1, \ldots, \widetilde{X}_M$ are conditionally independent given X^*. Also, for $1 \leq i \leq M$

$$\widetilde{X}_i \mid X^* \sim K^L(X^*, \cdot).$$

Now assume that $X_0 \sim \pi$. Since $\widehat{K}$ is the reversal of K, we know that $(X^*, X_0) \stackrel{d}{=} (Y_0, Y_1)$ where $Y_0 \sim \pi$ and $Y_1 \mid Y_0 \sim K^L(Y_0, \cdot)$. Thus.

$$X_0 \mid X^* \sim K^L(X^*, \cdot).$$

It follows that $X_0, \widetilde{X}_1, \ldots, \widetilde{X}_M$ are independent and identically distributed given X^*. Thus, $X_0, \widetilde{X}_1, \ldots, \widetilde{X}_M$ are exchangeable. □

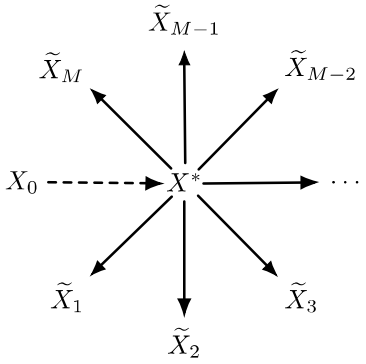


Figure 1: A graphical representation of the parallel method. The solid arrows represent running the Markov chain forwards for L steps and the dashed arrow represents running the Markov chain backwards for L steps.

By Proposition 3.3, the parallel method can be used to conduct valid MCMC tests. However, for small values of L, each sample $\widetilde{X}_i$ will be strongly dependent on X_0. This dependency means that the parallel method may be inconsistent.

Proposition 3.4. *Suppose $\widetilde{X}_1, \ldots, \widetilde{X}_M$ are generated via the parallel method. If the number of steps, L, is fixed and the number of samples, M, goes to infinity, then there exists a random variable p_∞ such that*

$$p_{MC} \xrightarrow{\mathbb{P}} p_\infty \overset{d}{=} K^L\left(X^*, \{x : T(x) \geq T(X_0)\}\right),$$

where $X^* \sim \widehat{K}^L(X_0, \cdot)$.

In words, p_{MC} converges to a mixture of analytic p-values. We first draw $X^* \sim \widehat{K}^L(X_0, \cdot)$. Next, we draw i.i.d. samples $\widetilde{X}_1, \ldots, \widetilde{X}_M \overset{\text{iid}}{\sim} K^L(X^*, \cdot)$. As M goes to infinity, p_{MC} will thus converge to the analytic p-value for the distribution $K^L(X^*, \cdot)$. Therefore, when using the parallel sampler, the limiting value of p_{MC} is random as it depends on the latent X^*. Some consequences of this are discussed in Section 4 where we use the parallel method to test the null hypothesis that X_0 is normally distributed with mean zero and variance one.

3.2 The permuted serial method

Our next exchangeable sampler is the *permuted serial method* [4, 2]. This method requires the same amount of computation as the parallel method but uses longer runs of the Markov chain to sample $\widetilde{X}_1, \ldots, \widetilde{X}_M$.

1. Sample a permutation σ uniformly from the set of all permutations of $\{0, 1, \ldots, M\}$.
2. Set $m^* = \sigma(0)$ and $Y_{m^*} = X_0$.
3. (a) From Y_{m^*} run a Markov chain according to $\widehat{K}^L$ for m^* steps to sample $Y_{m^*-1}, Y_{m^*-2}, \ldots, Y_0$.
   (b) From Y_{m^*} run a Markov chain according to K^L for $M - m^*$ steps to sample $Y_{m^*+1}, Y_{m^*+2}, \ldots, Y_M$.
4. For $i = 1, \ldots, M$, set $\widetilde{X}_i = Y_{\sigma(i)}$.

The permuted serial method is represented visually in Figure 2.

Remark. The original version of the serial method presented in [4] differs from the above procedure. Instead of sampling a permutation σ, [4] sample $m^* \sim \text{Unif}(\{0, \ldots, M\})$ and then continue to steps 2 and 3. They skipped step 4 and returned the p-value,

$$p = \frac{|\{0 \leq i \leq M : i \neq m^*, T(Y_i) \geq T(X_0)\}| + 1}{M + 1}.$$

The above quantity is invariant under permuting the samples $\{Y_i : i \neq m^*\}$. Thus, the original serial method produces the same p-value as the permuted version and the two are equivalent for testing purposes. However, only the permuted version produces an exchangeable sequence [2].

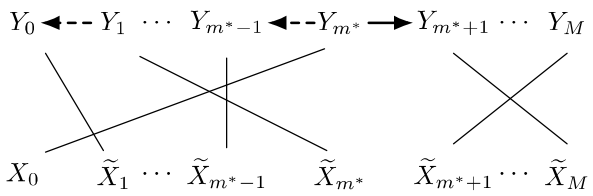


Figure 2: A graphical representation of the permuted serial method. Solid arrows represent running a Markov chain forwards for L steps and the dashed arrows represent running the Markov chain backwards for L steps. The lines represent the random permutation σ.

Proposition 3.5. *The permuted serial method is an exchangeable sampler.*

To prove that the permuted serial method is an exchangeable sampler, we will use the following lemma.

Lemma 3.6. *Let σ be the random permutation and $(Y_i)_{i=0}^M$ be the random samples from steps 1-3 of the permuted serial method. If $X_0 \sim \pi$, then $(Y_i)_{i=0}^M$ are independent of σ.*

Proof. Assume that $X_0 \sim \pi$. The random permutation σ only affects $(Y_i)_{i=0}^M$ through the random index $m^* = \sigma(0)$ at which we set $Y_{m^*} = X_0$. Thus, to show that $(Y_i)_{i=0}^M$ is independent of σ, we will show that the distribution of $(Y_i)_{i=0}^M$ does not depend on m^*.

To do this, recall that $Y_{m^*-1}, Y_{m^*-2}, \ldots, Y_0$ are generated by running a Markov chain from $Y_{m^*} = X_0$ according to $\widehat{K}^L$. By the assumption that $X_0 \sim \pi$ and the definition of $\widehat{K}$, this is equivalent to sampling $Y_0 \sim \pi$ and then running a chain from Y_0 according to K^L to sample $Y_1, Y_2, \ldots, Y_{m^*}$. The samples $Y_{m^*+1}, Y_{m^*+2}, \ldots, Y_M$ are generated by running a Markov chain from $Y_{m^*} = X_0$ according to K^L. Thus, the distribution of $(Y_i)_{i=0}^M$ is equivalent to sampling $Y_0 \sim \pi$ and running a Markov chain from Y_0 according to K^L to sample $Y_1, \ldots, Y_M$. This description of the distribution of $(Y_i)_{i=0}^M$ does not depend on m^*. Thus, $(Y_i)_{i=0}^M$ are independent of σ. □

We will now use Lemma 3.6 to prove that the permuted serial method produces exchangeable samples.

Proof of Proposition 3.5. Assume that $X_0 \sim \pi$ and let σ, $(Y_i)_{i=0}^M$ and $(\widetilde{X}_i)_{i=1}^M$ be as in the permuted serial method. By Lemma 3.6, $(Y_i)_{i=0}^M$ and σ are independent. Since $X_0 = Y_{\sigma(0)}$ and $\widetilde{X}_i = Y_{\sigma(i)}$, we can write the distribution of $X_0, \widetilde{X}_1, \ldots, \widetilde{X}_M$ in terms of the distribution of σ and $(Y_i)_{i=0}^M$. Specifically, if P_X is the distribution of $(X_0, \widetilde{X}_1, \ldots, \widetilde{X}_M)$ and P_Y is the distribution of $(Y_0, \ldots, Y_M)$, then

$$P_X(X_0, \widetilde{X}_1, \ldots, \widetilde{X}_M) = \sum_{\sigma \in S_{0:M}} \frac{P_Y\left(X_{\sigma^{-1}(0)}, \widetilde{X}_{\sigma^{-1}(1)}, \ldots, \widetilde{X}_{\sigma^{-1}(M)}\right)}{(M+1)!},$$

where $S_{0:M}$ is the set of permutations of $\{0, 1, \ldots, M\}$. The right-hand-side of the above equation is invariant under permuting $X_0, \widetilde{X}_1, \ldots, \widetilde{X}_M$. Thus, $X_0, \widetilde{X}_1, \ldots, \widetilde{X}_M$ are exchangeable. □

Under an additional assumption on the Markov chain K, the permuted serial method is consistent if we keep L fixed and send M to infinity. Intuitively, this is because the number of steps of the Markov chain between samples $\widetilde{X}_i$ tends to increases with M. The dependencies among $\widetilde{X}_1, \ldots, \widetilde{X}_M$ are thus decreasing with M. This in turn implies that a weak law of large numbers may be applied to the random variables $I[T(\widetilde{X}_i) \geq T(X_0)]$. More formally,

Proposition 3.7. *Assume that K^L is irreducible meaning that for all $x, y \in \mathcal{X}$, there exists j such that $k^{Lj}(x, y) > 0$. Suppose that $\widetilde{X}_1, \ldots, \widetilde{X}_M$ are sampled using the permuted serial method with L fixed. Then, as $M \to \infty$,*

$$p_{MC} \xrightarrow{\mathbb{P}} p_A.$$

Thus, the permuted serial method is consistent.

The proof of Proposition 3.7 is an application of the ergodic theorem for Markov chains (Theorem A.2). The details are given in the Appendix.

3.3 The permuted split-star method

Our next exchangeable sampler is a generalization of both the parallel method and the permuted serial method [8]. It can be motivated by noting that in Figure 1 each arrow from X^* to $\widetilde{X}_i$ represents an i.i.d. draw of a Markov chain that starts at X^* and is generated by K. This implies that the entire "spokes" in Figure 1 are exchangeable, not just the tips $\widetilde{X}_i$. We could therefore generate exchangeable samples by applying the permuted serial method to each of these spokes. This gives the exchangeable sampler called the *permuted split-star method* which was first described in [8].

The permuted split-star method is a more involved algorithm than the previous procedures. The intuitive idea is that we combine the parallel and series methods to produce samples $Y_0, Y_1, \ldots, Y_M$. This is done by creating M_2 spokes as in Figure 1 and then running the serial method to produce M_1 samples along each spoke as in Figure 2. Having done this, we uniformly permute $Y_0, Y_1, \ldots, Y_M$ to produce $X_0, \widetilde{X}_1, \ldots \widetilde{X}_M$. As in the permuted serial method, we use a combination of running the chain backwards and forwards so that the procedure is initialized at $X_0 = Y_{m^*}$ where m^* is uniform from $\{0, 1, \ldots, M\}$. The permuted split-star method proceeds as follows:

1. Write $M + 1 = M_1 M_2$.
2. Sample σ, a uniform permutation of $\{0, 1, \ldots, M\}$.
3. Write $\sigma(0) = M_1 m_2^* + m_1^*$ with $0 \le m_1^* < M_1$ and $0 \le m_2^* < M_2$.
4. Generate $Y_{M_1 m_2^*}, \ldots, Y_{M_1(m_2^*+1)-1}$ as in the step 3 of the permuted serial method with $m^* = m_1^*$ and $Y_{M_1 m_2^* + m_1^*} = X_0$.
5. From $Y_{M_1 m_2}$ run a Markov chain according to $\widehat{K}^L$ for one step to sample Y^*.
6. Independently, for $0 \le j < M_2$ and $j \ne m_2^*$, starting from Y^* run a Markov chain according to K^L for M_1 steps to sample $Y_{M_1 j+1}, \ldots, Y_{M_1(j+1)-1}$.
7. Return $\widetilde{X}_i = Y_{\sigma(i)}$ for $1 \le i \le M$.

Remark. In practice, it is not necessary to compute the full permutation σ. Instead, as in the permuted serial method, one can simply set $m_2^* = 0$ and sample m_1^* uniformly from $\{0, 1, \ldots, M_1 - 1\}$. Then, after completing steps 4-6 as above, we can return

$$p = \frac{|\{0 \le i \le M : i \ne m_1^*, T(Y_i) \ge T(X_0)\}| + 1}{M + 1}.$$

This quantity is equal to the p-value calculated using the exchangeable sample $X_0, \widetilde{X}_1, \ldots, \widetilde{X}_M$ but the samples $Y_0, Y_1, \ldots, Y_M$ will not be exchangeable. This procedure is the original *split-star method* proposed by [8].

The proof that the permuted split-star method is an exchangeable sampler is similar to the proof of Proposition 3.5. If $X_0 \sim \pi$ and we run the permuted split-star method, then, as in the permuted serial method, $(Y_i)_{i=0}^M$ are independent of σ. Since $X_0, \widetilde{X}_1, \ldots, \widetilde{X}_M$ is a uniform permutation of $(Y_i)_{i=0}^M$, we can conclude that $X_0, \widetilde{X}_1, \ldots, \widetilde{X}_M$ are exchangeable. To summarize,

Proposition 3.8. *The permuted split-star method is an exchangeable sampler.*

As claimed above, the parallel method and the permuted serial method are special instances of the permuted split-star method. Taking $M_1 = 1$ and $M_2 = M + 1$ gives the parallel method and taking $M_1 = M + 1$ and $M_2 = 1$ gives the permuted serial method.

Like the permuted serial method, the permuted split star method is consistent as M_1 goes to infinity under an irreducibility assumption. However, like the parallel method, the permuted split star method is not consistent if M_1 and L are fixed and M_2 goes to infinity.

3.4 Comparison

Each of the methods above have their advantages. The main benefit of the permuted serial method over the parallel method is its consistency. Even for finite M, we expect the permuted serial method to be closer to a standard Monte Carlo test than the parallel method. In the parallel method, each $\widetilde{X}_i$ is the result of running a Markov chain $2L$ steps from X_0. However, in the permuted serial method, the number of steps is $|\sigma(i) - \sigma(0)|L$ where σ is a uniformly drawn random permutation of $\{0, 1, \ldots, M\}$. Thus, $\widetilde{X}_i$ is an average of $\frac{M+2}{3}L$ steps away from X_0. In applications, the number of MCMC samples M will be in the hundreds or higher. Thus, the dependency between X_0 and $\widetilde{X}_i$ in the permuted serial method can be substantially less than in the parallel method. This decrease in dependency means that the permuted serial tests should behave more like a standard Monte Carlo test.

On the other hand, the parallel method has a computational advantage. All three methods require $M+1$ evaluations of the test statistic T and roughly ML steps of the Markov chain. However, as suggested by its name, the parallel method can be parallelized. Once the latent value X^* has been sampled, $\widetilde{X}_1, \ldots, \widetilde{X}_M$ can be computed in parallel. When parallel computation is available, we can run the parallel method with a much larger value of L. Increasing L decreases the dependency between samples. Thus, by parallelizing the computation of $\widetilde{X}_1, \ldots, \widetilde{X}_M$ and using a larger value of L, we can perhaps get an improvement over using the permuted serial method

The permuted split-star method combines the benefits of both the permuted serial and parallel methods. As noted in [8], the number of parallel processors available will typically be much less than the desired number of samples M. When using the permuted split-star method, we can set $M_2 - 1$ equal to the number of parallel processors. Step 6 of the permuted split-star method can then be done in parallel across these processors. Each processor will perform a version of the permuted serial method to produce M_1 samples. The permuted split-star method thus benefits from long runs of the Markov chain like the permuted serial method, and it can be parallelized like the parallel method [8].

4 Examples

In this section, we study two testings problems that highlight the difference between MCMC p-values and standard Monte Carlo p-values. In first example we examine the power of the parallel method when testing the mean of a normally distributed random variable with known variance. As in Section 6 of [4] we use an autoregressive Markov chain and examine how the choice of Markov chain affects power. In the second example, the null distribution is a bimodal mixture and the Markov chain is slow to mix. This example shows that there can be qualitative differences between MCMC tests and standard Monte Carlo tests.

4.1 Testing the standard normal distribution

Suppose our null hypothesis is $X_0 \sim \mathcal{N}(0, 1)$ where $\mathcal{N}(\mu, \tau^2)$ represents the normal distribution with mean μ and variance τ^2. There is no need to use MCMC to test this hypothesis, but it is illustrative to study these methods in this familiar setting. In Section 6 of [4], the authors study the power of the parallel test when using an autoregressive Markov chain to generate the MCMC samples $\widetilde{X}_1, \ldots, \widetilde{X}_M$. In this section we will derive a complementary result. As in Section 6 of [4], we will use the following Markov chain.

Definition 4.1. An order one autoregressive process with autocorrelation $\rho \in (-1, 1)$, is a Markov chain $(X_i)_{i=0}^M$ that transitions from X_i to X_{i+1} according to,

$$X_{i+1} = \rho X_i + \sqrt{1 - \rho^2} Z_i,$$

where $Z_i \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$. Equivalently, the Markov chain has a transition kernel K with density

$$k(x,y) = \varphi\left(y; \rho x, 1-\rho^2\right),$$

where $\varphi(\cdot;\mu,\tau^2)$ is the density of the normal distribution with mean μ and variance τ^2.

Autoregressive Markov chains are well-studied in time series models. For any $\rho \in (-1,1)$ the Markov chain in Definition 4.1 is reversible, and the standard normal distribution is the unique stationary distribution. However, for ρ close to 1, the correlation between samples is high, and the chain is slow to mix.

Suppose we use an order one autoregressive Markov chain in the parallel method to test $X_0 \sim \mathcal{N}(0,1)$. We will use the test statistic $T(x) = x$, so that large values of X_0 correspond to evidence against the null hypothesis. The analytic p-value is simply the tail probability $p_A = 1 - \Phi(X_0)$, where Φ is the cumulative distribution function for $\mathcal{N}(0,1)$. However, if we use the parallel method, then Proposition 3.4 implies the following.

Proposition 4.2. *Suppose that $\widetilde{X}_1, \ldots, \widetilde{X}_M$ are sampled by using the parallel method with an order one autoregressive Markov chain with correlation ρ. If the number of steps is fixed at L and the number of samples M goes to infinity, then*

$$p_{MC} \xrightarrow{\mathbb{P}} p_\infty \overset{d}{=} 1 - \Phi\left(\sqrt{1-\rho^{2L}} X_0 - \rho^L Z^*\right),$$

where $Z^* \sim \mathcal{N}(0,1)$ and Z^* is independent of X_0.

Note that the limiting distribution depends only on ρ^L and not on ρ or L individually. This is expected since running a chain with autocorrelation ρ for L steps is equivalent to running a chain with autocorrelation ρ^L for one step.

From Proposition 4.2, we can study the limiting power of the parallel method. Specifically, consider the alternative $X_0 \sim \mathcal{N}(\mu,1)$ with $\mu > 0$. The uniformly most powerful level α test for this alternative rejects the null when $X_0 \geq \Phi^{-1}(1-\alpha)$ or equivalently when $p_A \leq \alpha$. The power of this analytic test is $1 - \Phi(\Phi^{-1}(1-\alpha) - \mu)$. On the other hand, if we reject the null when $p_\infty \leq \alpha$, then our power is

$$1 - \Phi(\Phi^{-1}(1-\alpha) - \sqrt{1-\rho^{2L}}\mu). \tag{3}$$

Thus, as shown in [4], the dependencies in a MCMC test reduce the signal strength, μ, by a factor of $\sqrt{1-\rho^{2L}}$. For ρ^L close to 1, this loss in power can be substantial as shown in Figure 3.

This example also shows the importance of choosing the Markov chain K. Here K is determined by the choice of the autocorrelation ρ, but in other situations there could be a variety of Markov chains available. The results in this section suggest that Markov chains with faster mixing times will produce more powerful tests. Thus, one can use standard diagnostics from applied Bayesian statistics such as trace plots and effective sample sizes to assess the mixing time. These methods can also be used to pick the parameter L. However, for the MCMC samples to remain valid all diagnostics must be independent of the data X_0.

4.2 Testing a bimodal distribution

Let $\mathcal{X}$ be the discrete set $\{1, 2, \ldots, 100\}$ and define π by

$$\pi(x) \propto \frac{1}{2}\varphi\left(x; 25, 6^2\right) + \frac{1}{2}\varphi\left(x; 75, 6^2\right),$$

where again $\varphi(\cdot;\mu,\tau^2)$ is the density of the normal distribution with mean μ and variance τ^2. The distribution π has two modes, one at $x = 25$ and another at $x = 75$. As before, let $T(x) = x$ so that large values of X_0 correspond to evidence against the hypothesis $X_0 \sim \pi$. Again the analytic p-value is the tail probability

$$p_A = \sum_{x \geq X_0} \pi(x).$$

The analytic p-value is a decreasing function of X_0 and $p_A \leq 0.05$ if and only if $X_0 \geq 83$. The density $\pi(x)$ and the rejection region $\{X_0 \geq 83\}$ are shown in Figure 4.

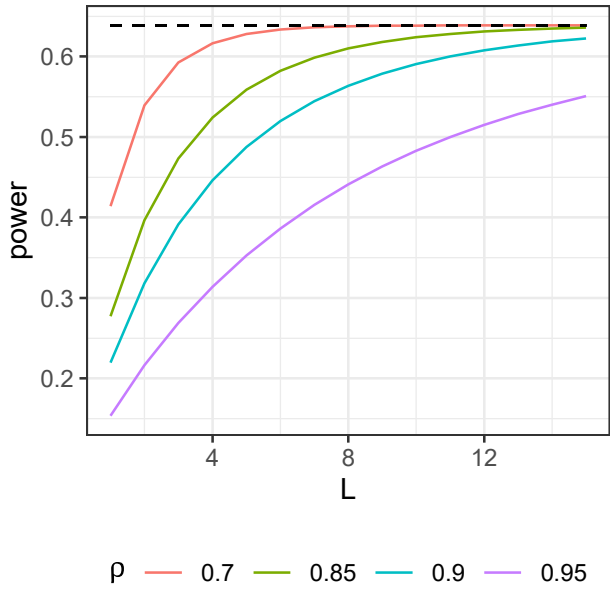


Figure 3: The limiting power of the parallel method from (3) is shown as a function of L for $\mu = 2$, $\alpha = 0.05$ and various values of ρ. The dashed line in black is $1 - \Phi(\Phi^{-1}(0.95) - 2)$, the highest possible power at level α. When $\rho = 0.7$, the power of the Monte Carlo test quickly approaches the optimal power. When ρ is close to 1, this convergence is slow.

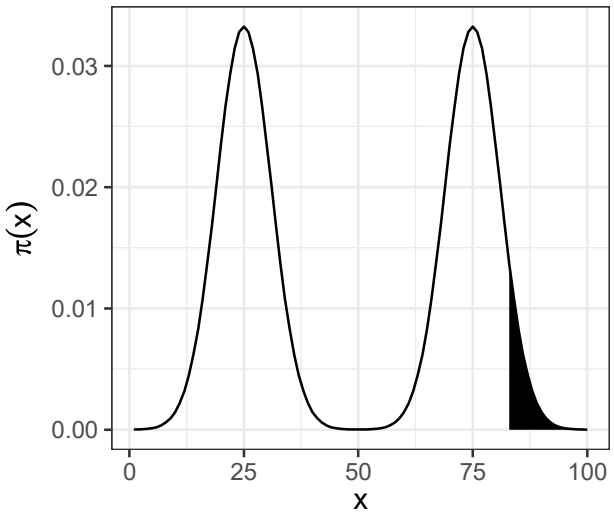


Figure 4: The density π has modes at $x = 25$ and $x = 75$. The shaded region shows the 95th percentile of π.

We will use Metropolis–Hastings to construct a reversible Markov chain with stationary distribution π. Our proposal distribution will be $x' = x \pm 1$ with equal probability. Applying Metropolis–Hastings to this proposal produces a reversible transition kernel K with stationary distribution π. If $a(x, x') = \min\{1, \pi(x')/\pi(x)\}$, then transition kernel K is given by

$$x \mapsto y = \begin{cases} x+1 & \text{w.p. } \frac{a(x,x+1)}{2}, \\ x-1 & \text{w.p. } \frac{a(x,x-1)}{2}, \\ x & \text{w.p. } 1 - \frac{a(x,x+1)+a(x,x-1)}{2}. \end{cases}$$

Since π is bimodal and our proposal distribution takes small steps, K will be very slow to mix. Specifically, a Markov chain initialized near one mode will take a very long time to transition to a point near the other mode. Thus, when using K to perform an MCMC test, we would expect $\widetilde{X}_1, \ldots, \widetilde{X}_M$ to be highly dependent on which mode X_0 is closer to. For example, if $X_0 = 40$, then it is very unlikely that any of $\widetilde{X}_1, \ldots, \widetilde{X}_M$ will lie in $\{50, \ldots, 100\}$. The MCMC tests are thus likely to reject the null hypothesis when $X_0 = 40$ whereas a standard Monte Carlo test essentially never rejects when $X_0 = 40$.

This intuition is confirmed by a simulation whose results are summarized in Table 1. In the simulation we drew $X_0 \sim \pi$ 2500 times and computed p_{MC} using different samplers. The bottom row reports the percentage of simulations when $p_{MC} \leq 0.05$. This is the proportion of times we would reject $X_0 \sim \pi$. By the validity of these sampler, each entry in bottom row should be at most 5%. The first and second rows show the percentage of simulations when $p_{MC} \leq 0.05$ and $X_0 \leq 50$ and $X_0 > 50$ respectively. The standard Monte Carlo test does not reject when $X_0 \leq 50$ since the rejection region for the analytic p-value is $\{X_0 \geq 83\}$. On the other hand, the MCMC tests reject when X_0 is in the right tail of either mode. For both MCMC methods, the percentage of rejections when $X_0 \leq 50$ is close to the percentage of rejections when $X_0 > 50$. By marginal validity, these two percentages should be close to 2.5%.

Table 1: MCMC p-values and standard Monte Carlo p-values were calculated for 2500 values of $X_0 \sim \pi$. The table reports the percentage of simulations for which $p_{MC} \leq 0.05$. For all methods $M = 99$ samples were taken. For the MCMC methods, $L = 100$ steps were used. Code to reproduce this table and the other examples can be found at https://github.com/Michael-Howes/MCMC-significance-tests.

	Standard	Parallel	Permuted Serial
$X_0 \leq 50$	0%	2.4%	2.6%
$X_0 > 50$	4.4%	2.2%	2.0%
	4.4%	4.6%	4.6%

In general, if we use an MCMC test with a slow to mix Markov chain, then we should expect the MCMC test to behave very differently to a standard Monte Carlo test. When we compare X_0 to $\widetilde{X}_1, \ldots, \widetilde{X}_M$ in a standard Monte Carlo test, we are measuring how "globally extreme" X_0 is. This is because $T(X_0)$ is compared to $T(\widetilde{X})$ for $\widetilde{X}$ a typical sample from π. On the other hand, in an MCMC test, we are measuring how "locally extreme" X_0 is. We are comparing $T(X_0)$ to $T(\widetilde{X})$ where $\widetilde{X}$ depends on X_0. When the Markov chain K is slow to mix, these two forms of "extreme" may differ substantially as the current example shows.

The fact that MCMC tests reject when X_0 is locally extreme can be an advantage. In this example, we see that the MCMC tests have power against alternatives that put high probability on the interval $\{40, \ldots, 50\}$ whereas the standard test has no power against these alternatives. Further consequences of the difference between locally and globally extreme values of X_0 are explored in [9, 10, 11] where MCMC methods are used to detect gerrymandering as discussed in Section 5.2.

5 Applications

Since their appearance in [4], MCMC significance tests have been applied in many domains [2, 3, 9, 8, 16, 17]. Here we will discuss some of these applications.

5.1 The Rasch model

The Rasch model is a model for binary matrices used for item-response data. Suppose I subjects answer a survey with J yes/no questions. The results from the survey can be encoded in an $I \times J$ binary matrix A with $A_{ij} = 1$ if and only if subject i answered yes to question j. The Rasch model assumes that the entries A_{ij} are independent Bernoulli random variables with

$$\mathbb{P}_{\beta,\gamma}(A_{ij} = 1) = \frac{\exp(\beta_i - \gamma_j)}{1 + \exp(\beta_i - \gamma_j)}. \tag{4}$$

The model has $I+J-1$ free parameters. When the survey questions are tests, the parameter β_i is interpreted as subject i's ability and γ_j is interpreted as question j's difficulty. This model is an exponential family where the sufficient statistics are the row and column sums of A.

Suppose that we have a binary matrix $X_0 \in \{0,1\}^{I\times J}$, and we want to know if (4) is an appropriate model for X_0. That is, we want to test the composite hypothesis $X_0 \sim \mathbb{P}_{\beta,\gamma}$ for some unspecified (β, γ). This composite hypothesis can be turned into a single hypothesis by conditioning on the row and column sums. More precisely, let R_0 and C_0 be the row and column sums of X_0 and define $\mathcal{X}$ to be the set of all binary matrices with row sums R_0 and column sums C_0. If $X_0 \sim \mathbb{P}_{\beta,\gamma}$ for any (β, γ), then $X_0 \mid R_0, C_0 \sim \pi$ where π is the uniform distribution on $\mathcal{X}$. Therefore, testing $X_0 \mid R_0, C_0 \sim \pi$ is a conditional test of the composite hypothesis $X_0 \sim \mathbb{P}_{\beta,\gamma}$ for some (β, γ).

For even a moderate number of items and subjects, sampling directly from the uniform distribution on $\mathcal{X}$ is intractable. A Markov chain with stationary distribution π is given in [4] and a sequential importance sampling approach is described in [7]. More recently, a new Markov chain called the rectangle loop algorithm was defined in [18]. The rectangle loop algorithm has π as its only stationary distribution and was shown to sample more efficiently than the Markov chain in [4]. Here we use the rectangle loop algorithm with the permuted serial method.

We used the data set `verbal` from the R package `difR` [14]. The data set consists of $I = 316$ subjects, each of whom answered $J = 24$ questions about verbal aggression. Each question involved reading the description of a frustrating situation. The subjects then answered whether they would respond in a certain way such as swearing. The data set also contains each subject's gender. The test statistic used was Andersen's likelihood ratio test [1]. This statistic is based on the conditional likelihood,

$$L(\gamma) = \mathbb{P}_\gamma \left(X_0 \mid R_0 \right).$$

The conditional likelihood does not depend on β since the row sums R_0 are sufficient for β. Andersen's likelihood ratio equal to the log conditional likelihood from fitting a separate Rasch model for each gender minus the log conditional likelihood from fitting a single Rasch model. The statistics were computed using the function `LRtest` from the R package `eRm` [15]. The results are summarized in Figure 5 which shows strong evidence against the single Rasch model. This suggests that the genders differ in their relative responses to different situations.

The Rasch model is also used in ecology [7]. Binary matrices $A \in \{0,1\}^{I\times J}$ can be used to record the presence and absence of species at different locations. The rows represent different species and the columns represent different locations. A value of $A_{ij} = 1$ means that species i is present at location j. If X_0 is the observed presence/absence matrix for a group of species and locations, then conditioning on the row and column sums of X_0 is one way of controlling for species abundance and location habitability. The hypothesis that X_0 is drawn from the uniform distribution π implies that there are no interactions between species. Statistics such as the number of instances of two species living at the same location can be used to measure the interactions between species. MCMC significance tests can then be used to calibrate these test statistics by producing valid p-values.

5.2 Political bias in district maps

In [9] and [8], the authors use MCMC methods to test whether a district map has a political bias. Here a district is defined as collection of census blocks. The number of districts is considered fixed, and a district map is an assignment of each census block to one of the districts. A district map is *valid* if each district is

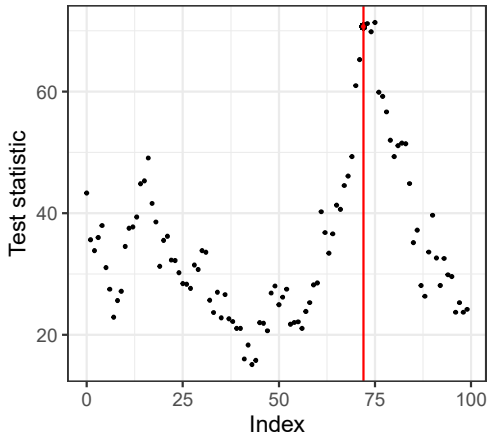


Figure 5: The results from using the serial method to perform a goodness of fit test for the Rasch model. The red point equals the value of the test statistic on the observed data X_0. The red line is show the random index m^* where we have $Y_{m^*} = X_0$. The black points are equal to the comparison values $T(\widetilde{X}_i)$. To highlight the dependency between the samples $\widetilde{X}_1, \ldots, \widetilde{X}_M$, we have plotted the test statistics in their order prior to being permuted. The test statistic evaluated on the true data, $T(X_0)$ is third largest amongst $T(X_0), T(\widetilde{X}_1), \ldots, T(\widetilde{X}_M)$. This corresponds to a p-value of 0.03 giving evidence against the Rasch model.

approximately equal in population and the districts are what the authors call "compact" which means each district is not too irregular.

The authors of [9] analyzed the district map of Pennsylvania in 2012. In 2012, Pennsylvania contained 18 districts and roughly 9,000 census blocks. Using our notation, $\mathcal{X}$ is equal to the set of all ways of assigning these 9,000 census blocks to the 18 districts such that the resulting district map is valid. Our observed data X_0 is the district map used in 2012 election and the null distribution, π, is the uniform distribution on $\mathcal{X}$.

Sampling directly from π is impossible. The authors instead construct a Markov chain that uniformly chooses a census block on the border of two districts. The Markov chain then proposes swapping the assignment of the chosen census block. If the swapped block results in a valid district map, then the proposal is accepted. Otherwise, the state of the Markov chain does not change. By adding a Metropolis–Hastings step, the authors produce a Markov chain on space of valid district maps which has π as a stationary distribution. The Markov chain is also shown to be reversible, so that $\widehat{K} = K$. Figure 6 is from [9] and shows the 2012 district map of Pennsylvania and a district map produced by taking 2^{40} Markov chain steps.

In [8] the authors use a method called the $\sqrt{\varepsilon}$-test to test the hypothesis that the current district map is drawn from the uniform distribution π. This method is based on the following bound.

Theorem 5.1 (Theorem 1.1 in [9]). *Let K be a reversible transition kernel with stationary distribution π. Suppose that $X_0, \widetilde{X}_1, \ldots, \widetilde{X}_M$ is a Markov chain initialized at X_0 with transition kernel K^L. Then $\sqrt{2p_{MC}}$ is a valid p-value under the hypothesis $X_0 \sim \pi$.*

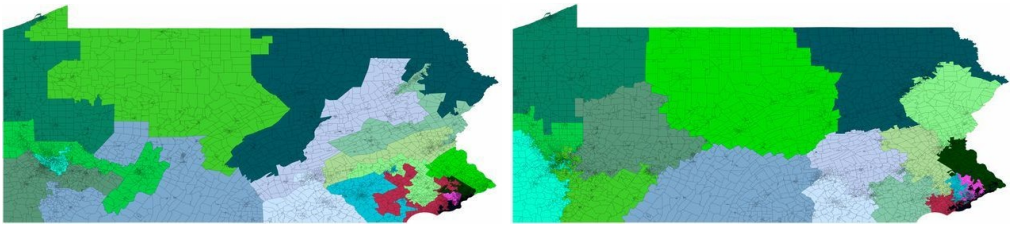


Figure 6: (Left) The 2012 district map of Pennsylvania. (right) The district map produced by running a Markov chain for 2^{40} steps. Figure from [9].

Theorem 5.1 offers an alternative to exchangeable samplers. Instead of generating exchangeable samples, one can directly use the sequential samples from running a single Markov chain. The $\sqrt{\varepsilon}$-test is thus simpler and easier to interpret than the previous tests based on exchangeability. However, since $p_{MC} \in [0,1]$, we have $\sqrt{2p_{MC}} > p_{MC}$. Thus, the $\sqrt{\varepsilon}$-test will typically have less power than the methods in Section 3.

Using either the $\sqrt{\varepsilon}$-test or an exchangeable sampler requires choosing a test statistic T. The test statistic would ideally give tests that have power against "political motivated" alternative distributions. That is, distributions that put high probability on district maps that give a partisan advantage. The authors thus construct test statistics based on voting data from the 2012 Pennsylvania election. For each census block, the authors have the number of votes for the Democratic and Republican parties from the 2012 election. Therefore, given a district map, they can calculate what the election result would have been with that district map. Formally, for each district j, there is a function D_j from $\mathcal{X}$ to the interval $[0,1]$ where for a given district map $x \in \mathcal{X}$, $D_j(x)$ is the proportion of votes for the Democratic Party in district j. From here, the authors consider a number of statistics such as the variance of the vector $D(x)$ and the difference between the mean and the median of $D(x)$. The authors find highly significant results for all the statistics they consider. One of the authors testified in a state supreme court case that found the 2012 Pennsylvania district map to be unconstitutional [9].

5.3 Goodness of fit testing

In section 5.1 we tested the composite null that our observed table was distributed according to a Rasch model with unknown parameters. This was done by conditioning on a sufficient statistic to turn the composite hypothesis into a simple hypothesis. A similar procedure can be used when the null hypothesis is a discrete exponential family. Suppose $\mathcal{X}$ is a finite set, and we have a family of distributions $\mathbb{P}_\eta$ given by,

$$\mathbb{P}_\eta(x) = \exp\left(\sum_{j=1}^{d} S_j(x)\eta_j - A(\eta)\right), \tag{5}$$

where $S_j : \mathcal{X} \to \mathbb{R}$ are functions, $\eta \in \Omega \subseteq \mathbb{R}^d$ are the natural parameters and $A(\eta)$ is a log normalizing constant. The function S_j are sufficient statistics for η and if $X \sim \mathbb{P}_\eta$, then $X \mid (S_j(X))_{j=1}^d = s$ is uniform on the set

$$\mathcal{X}_s = \{x \in \mathcal{X} : S_j(x) = s_j \text{ for } 1 \le j \le d\}.$$

Testing $X_0 \sim \mathbb{P}_\eta$ for some η has thus been reduced to testing $X_0 \mid (S_j(X_0))_{j=1}^d = s \sim \pi$ where π is the uniform distribution on $\mathcal{X}_s$. In [12], a general construction is given for finding connected Markov chains on $\mathcal{X}_s$ which converge to a given distribution. These chains can be used in an exchangeable sampler to perform goodness of fit tests for any parametric family of the form (5).

In a different direction, [2] propose a goodness-of-fit test based on *approximate sufficiency*. There, the authors also wish to test a parametric model $X_0 \sim \mathbb{P}_\eta$ for some $\eta \in \Omega \subseteq \mathbb{R}^d$. They use an approach related to the parametric bootstrap [13]. Using the parametric bootstrap one would sample $\widetilde{X}_1, \ldots, \widetilde{X}_M \overset{\text{iid}}{\sim} \mathbb{P}_{\hat{\eta}}$ where $\hat{\eta} = \hat{\eta}(X_0)$ is the maximum likelihood estimate (MLE) given the data X_0. The problem with this is that $X_0, \widetilde{X}_1, \ldots, \widetilde{X}_M$ are not exchangeable under the null. To create a valid method, the authors sample $\widetilde{X}_1, \ldots, \widetilde{X}_M$ from the conditional distribution

$$\mathbb{P}_{\hat{\eta}'(X_0)}(X \mid \hat{\eta}'(X) = \hat{\eta}'(X_0)), \tag{6}$$

where $\hat{\eta}'(X)$ is a maximizer of

$$\ell(\eta) = \log \mathbb{P}_\eta(X) + \eta^\top U, \quad U \sim \mathcal{N}(0, \tau^2 I_d).$$

That is, $\hat{\eta}'(X)$ is a noisy version of the MLE $\hat{\eta}(X)$. This noisy MLE, $\hat{\eta}'(X)$ is used instead of the true MLE $\hat{\eta}(X)$ because in some models, the conditional distribution $X \mid \hat{\eta}(X)$ is degenerate. By adding noise to $\hat{\eta}(X)$ we are essentially conditioning on less information about X. The conditional distribution $X \mid \hat{\eta}'(X)$ is thus more variable than $X \mid \hat{\eta}(X)$.

In [2], the authors compute the conditional distribution (6) up to a normalizing constant and derive a Markov chain with (6) as a stationary distribution. They also show that if $X_0 \sim \mathbb{P}_{\eta_0}$ for some $\eta_0 \in \Omega$, then

$$\mathbb{P}_{\hat{\eta}'(X_0)}(X|\hat{\eta}'(X) = \hat{\eta}'(X_0)) \approx \mathbb{P}_{\eta_0}(X|\hat{\eta}'(X) = \hat{\eta}'(X_0)).$$

The upshot is that if $\widetilde{X}_1, \ldots, \widetilde{X}_M$ are sampled from an exchangeable sampler with stationary distribution (6), then $X_0, \widetilde{X}_1, \ldots, \widetilde{X}_M$ are approximately exchangeable under the null. The authors make this approximation precise and show that their method produces asymptotically valid p-values for goodness of fit testing in a variety of parametric models.

5.4 The conditional permutation test

In [3], the authors develop a test of conditional independence for "semi-supervised" statistical problems. These are statistical problem where we have two data sets, a small labelled data set $(X_j, Y_j, Z_j)_{j=1}^n$, and a large unlabeled data set $(X'_j, Z'_j)_{j=1}^N$. In [3], it is assumed that there is enough unlabeled data so that the conditional distribution of $X \mid Z = z$ can be estimated precisely and treated as known. The goal is to test the hypothesis that Y is conditionally independent of X given Z. That is, the null hypothesis is

$$(X_j, Y_j, Z_j) \overset{\text{iid}}{\sim} P,$$

where P is any distribution that factors as

$$P(x, y, z) = Q_{X|Z}(x \mid z)Q_{Y|Z}(y \mid z)Q_Z(z). \tag{7}$$

The factorization (7) is equivalent to the statement that X and Y are conditionally independent given Z. As stated above, it is assumed that $Q_{X|Z}$ is known, but no assumptions are made on $Q_{Y|Z}$ or Q_Z. These are the same assumptions made of the *conditional randomization test (CRT)* [6]. The CRT is a standard Monte Carlo test that conditions on $(Y_j, Z_j)_{j=1}^n$ and resamples X_j. More precisely, let $D_0 = (X_j, Y_j, Z_j)_{j=1}^n$ be our observed data set. For $i = 1, \ldots, M$ and $j = 1, \ldots, n$, independently sample

$$\widetilde{X}_j^{(i)} \sim Q_{X|Z}(\cdot \mid Z_j), \tag{8}$$

and define $\widetilde{D}_i = (\widetilde{X}_j^{(i)}, Z_j, Y_j)_{j=1}^n$. Under the conditional independence assumption (7), $D_0, \widetilde{D}_1, \ldots, \widetilde{D}_M$ are an i.i.d. sample from the conditional distribution of D_0 given $(Z_j, Y_j)_{j=1}^n$.

The conditional permutation test (CPT) [3] is a variant of the above procedure. In addition to conditioning on $(Z_j, Y_j)_{j=1}^n$, the CPT also conditions on the multi-set $\{X_j\}_{j=1}^n$. This means that when sampling in (8), $\widetilde{X}_1^{(i)}, \ldots, \widetilde{X}_n^{(i)}$ must be a permutation of the observed data $X_1, \ldots, X_n$. Thus, instead of sampling $\widetilde{X}_1^{(i)}, \ldots, \widetilde{X}_n^{(i)}$ we can sample a permutation $\sigma^{(i)}$ and set

$$\widetilde{X}_j^{(i)} = X_{\sigma^{(i)}(j)}, \quad \widetilde{D}_i = \left(\widetilde{X}_j^{(i)}, Y_j, Z_j\right)_{j=1}^n.$$

For $\widetilde{D}_i$ to be drawn from the same conditional distribution as $D_0 \mid (Z_j, Y_j)_{j=1}^n, \{X_j\}_{j=1}^n$ we need that the permutation $\sigma^{(i)}$ is drawn from the distribution

$$P(\sigma) = \frac{\prod_{j=1}^n Q(X_{\sigma(j)} \mid Z_j)}{\sum_{\sigma'} \prod_{j=1}^n Q(X_{\sigma'(j)} \mid Z_j)}, \tag{9}$$

where the sum in the denominator is over all permutations of $\{1, \ldots, n\}$. The computational cost in evaluating such a sum grows quickly with n. To avoid this computational cost, the authors construct a Markov chain on permutations of length n with stationary distribution (9). They then use the parallel method to generate samples $\widetilde{D}_1, \ldots, \widetilde{D}_M$ that are exchangeable under the null hypothesis that X and Y are conditionally independent given Z. The authors find that the CPT has power comparable to that of the CRT and is more robust against misspecification of the conditional distribution $Q_{X|Z}(\cdot \mid z)$

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Supplementary material

Code to reproduce the examples and simulations from this paper is available at https://github.com/Michael-Howes/MCMC-significance-tests/

A Consistency of the permuted serial method

Here we prove Proposition 3.7. First recall that an irreducible Markov chain is *positive recurrent* if the expected return time to any state $x \in \mathcal{X}$ is finite. To prove Proposition 3.7, we will use the following two theorems.

Theorem A.1 (Theorem 3.2.6 in [5]). *An irreducible Markov chain is positive recurrent if and only if there exists a stationary distribution. Moreover, the stationary distribution π is, when it exists, unique and $\pi(x) > 0$ for all x.*

Theorem A.2 (Theorem 3.3.2 in [5]). *Let $(Y_i)_{i=0}^M$ be an irreducible positive recurrent Markov chain with stationary distribution π. Let $g : \mathcal{X} \to \mathbb{R}$ be any measurable function with*

$$\mathbb{E}_\pi[|g(Y)|] < \infty.$$

Then for any initial distribution $Y_0 \sim \pi'$ as $M \to \infty$

$$\frac{1}{M}\sum_{i=1}^{M} g(Y_i) \xrightarrow{\mathbb{P}} \mathbb{E}_\pi[g(Y)].$$

Proof of Proposition 3.7. Let $m^* \in \{0, \dots, M\}$, $(Y_i)_{i=0}^M$ and $(\widetilde{X}_i)_{i=0}^M$ be as in the permuted serial method. Note that,

$$\begin{aligned} p_{MC} =& \frac{1}{M+1}\left(\sum_{i=1}^{M} I[T(\widetilde{X}_i) \geq T(X_0)] + 1\right) \\ =& \frac{1}{M+1}\sum_{i=0}^{m^*-1} I[T(Y_i) \geq T(X_0)] + \frac{1}{M+1} + \frac{1}{M+1}\sum_{i=m^*+1}^{M} I[T(Y_i) \geq T(X_0)] \end{aligned}$$

Since $\frac{1}{M+1} \to 0$ and $\frac{M}{M+1} \to 1$ as $M \to \infty$, it suffices to show the consistency of

$$p'_{MC} = \frac{1}{M}\sum_{i=0}^{m^*-1} I[T(Y_i) \geq T(X_0)] + \frac{1}{M}\sum_{i=m^*+1}^{M} I[T(Y_i) \geq T(X_0)]. \tag{10}$$

We have assumed that K^L is irreducible and that π is a stationary distribution for K. This implies that K^L and $\hat{K}^L$ are both irreducible and have stationary distribution π. Theorem A.1 thus implies that K^L and $\widehat{K}^L$ are both irreducible positive recurrent. We can therefore apply Theorem A.2 with $g(y) = I[T(y) \geq T(X_0)]$ to both K^L and $\widehat{K}^L$. If $Y_0 = X_0, Y_1, \dots, Y_m$ is a Markov chain with transition kernel K^L or $\widehat{K}^L$, then, as $m \to \infty$,

$$\begin{aligned} \frac{1}{m}\sum_{i=1}^{m} I[T(Y_i) \geq T(X_0)] &\xrightarrow{\mathbb{P}} \pi(\{y : T(y) \geq T(X_0)\}) \\ &= p_A. \end{aligned}$$

In (10), $Y_{m^*-1}, \ldots, Y_0$ are generated by running a Markov chain according to $\widehat{K}^L$ from X_0 and likewise $Y_{m^*+1}, \ldots, Y_M$ are generated by running a Markov chain according to K^L from X_0. Suppose now that we fix $\varepsilon > 0$. As M goes to infinity, m^* and $M - m^*$ both go to infinity with probability one. Thus, as $M \to \infty$, we will have that

$$p_1 = \frac{1}{m^*} \sum_{i=1}^{m^*-1} I[T(Y_i) \geq T(X_0)], \text{ and } p_2 = \frac{1}{M - m^*} \sum_{i=m^*+1}^{M} I[T(Y_i) \geq T(X_0)],$$

are within ε of p_A with probability tending to one. Since p'_{MC} is convex combination of p_1 and p_2, we can conclude that p'_{MC} will also be within ε of p_A with probability tending to one. □

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